



Self-repairing shoes and tires

Researchers at the University of Southern California have developed a 3D-printed rubber materials that self-repairs. <https://digit.in/mar19-17>



AI to boost Earth system science

New study shows that AI can substantially improve our understanding of the climate and the Earth system. <https://digit.in/mar19-18>

When it comes to physically created 2D sheets, the tape approach is still used and is the best method around, but this is not feasible for large scale production. Additionally, peeling off the tape creates cracks in the sheets. Creating 2D sheets is an area of active research. The single layer sheets can be “grown” on hot conducting plates, or the molecules are exfoliated in solvents. One of the cutting edge approaches, developed by researchers at MIT in late 2018, involves growing thick stacks of the 2D materials on sapphire crystals, then applying a thin nickel film on top. The 2D materials stick to the nickel more than the sapphire, and lifting the nickel film separates the 2D sheets from the sapphire wafer. Then, another nickel plate is added below the stack, which splits up the number of layers in the stack. Repeating the process results in a single sheet attached to the nickel film. This layer can then be deposited on the substrate for electronics. The technique allows for harvesting wafers of 2D materials, 2 inches in diameter each. The approach works for a number of 2D materials, including graphene, hexagonal boron nitride, tungsten disulfide, and molybdenum disulfide.

Another approach called chemical vapor deposition (CVD), which allows for the creation of 2D sheets of any size. The method was demonstrated for a 2D material called molybdenum telluride, where tellurium powder was vaporised, and deposited on a layer of pure molybdenum. The size of the sheets are limited only by the dimensions of the chamber in which the vaporisation occurs. Researchers are trying to create other 2D materials using this method.

APPLICATIONS

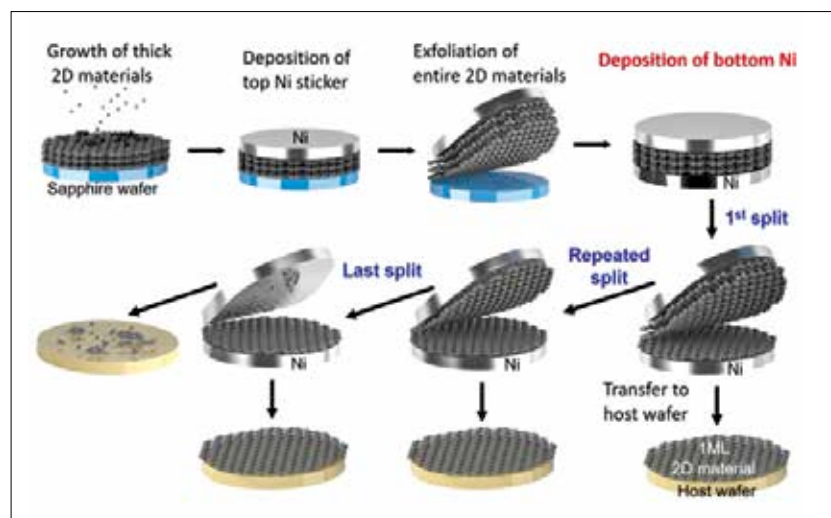
The unique properties and versatility of 2D materials mean that they have widespread potential applications. One of the areas that stand to benefit humans, is in medicine. The lattice structure of these materials are ideal for embedding drugs, and delivering them to targeted locations within the body. The large surface area makes the materials idea for use as sensors, which can monitor glucose, haemoglobin or cholesterol in real time,

and with precision. The materials may be engineered into variants that can attack cancer cells or act as antibiotics. Outside the body, the strength to weight ratio of the materials make them suitable for treating wounds. Graphene could also be potentially used in the process of tissue regeneration. There is a need for researching the biocompatibility of 2D materials, after which any medical applications would have to go through a rigorous testing process, before getting on to the market. 2D materials are expected to be widely used in biomedical applications by the 2030s.

2D materials can be integrated with existing materials, to create what are called composite materials. Currently,

rust over time. The same principle can be applied in food packaging as well, preventing foodstuff from going bad. If 2D materials are mixed with concrete which is then used in construction, architects and engineers can reduce the weight of buildings, allowing them to be taller. One of the use cases where graphene based materials is already seeing, is that of sports equipment. There are graphene based tennis racquets already in the market. In the near future, graphene based skis and bicycle tyres are expected.

The properties of 2D materials also make them ideal for use in electronics. They can be used to make stronger touchscreens that are more resilient



The process of depositing a 2D material on top of a silicon wafer

Credit: MIT

carbon fibre is being incorporated in aerospace engineering, because of its strength and lightness. Graphene is stronger and lighter, and could be used in aircraft construction in the future, which would make the aircraft lighter, and therefore more fuel efficient, with an increase in range as a bonus. The same material could protect the aircraft from lightning strikes, as well as act as a sensor that informs the pilots of excessive stresses in any part of the craft. The approach could be used to build ships and cars as well, which would create longer lasting vehicles. The use of graphene to isolate water and oxygen could mean that the self driving cars of the future will not deteriorate or

to cracking. The sensitivity of sensors based on 2D materials could reduce errors in touch interfaces, and significantly improve the user experience. Electronics can be further miniaturised with 2D materials, and be used to make faster electronic circuitry. The world's smallest transistor, is made from graphene and is made up of just ten atoms. Finally, the flexibility of 2D materials allows them to be used in wearable devices such as fitness trackers or smartwatches, as well as technology integrated into items of clothing. In the future you can potentially be reading this article on a flexible screen, that has the same appearance and texture as a sheet of paper. 